

Screenshot 2025-07-14 074052

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geflection" This phenomenn occurs #8 8 FN} ea inteinal voflectisnof the incident Wght'in

Tee cet rin around ls specu elector angle. Due te Ws phetameroy, the

orface appears to bent is oignal clara whiteole,

'Diffuse reflection'o' dull 'ita slftocls the light 6 ali dleections. Spesiiianrellsetion cous

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214 Computer Graphics

mirror, In contrasat fo diffuse reflection, ideal specular -teflection takes. place only in one inection. The difference between diffuse and specular reflection fsilustrate in fig betow.

Sporvlat relection

Figure 99: Difference between diffuse and specular reflections

Shiny surfaces very often have.a very thin transparent layer, for instance vamish. When light 'hts the suafece, part of the light penetrates the varnish layer anc is reflected orf the dull surface 'ofthe object. This part of the light is subject to diffuse reflection-arid the coloy'of the reflected light depends strongly on the ground color of the dull surface, Another part of the Light is* disecly reflected on the transparent layer by specular ieflection, Therefore, specular reflection does usually not change the'tolot 6f the Hight! The position of the viewer is of importance for calculating effects coming from diffuse reflectidn, Whether or Where the viewer can see specular

'The Phong specular reflection wioddI described by the relation, =.

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'Where, h is the intensity of the light' The value $0 \leq w(\theta) \leq 1$ is the fraction of the light which is

directly reflected at the shiny surface; n is the specular reflection exponent of the surface, For a
(perfect mirror, $n \rightarrow \infty$ would hold, A smaller n leads to less focused specular reflection,